From Behavior Cloning to Offline Reinforcement Learning

Learning Better Decisions from Logged Data

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From Prediction to Decision

Prediction	Decision
What will happen?	What should we do?
Truck arrival time	Which truck to unload next
Grain quality	Where to route the grain
Train delay	Which loading schedule to use

Machine learning describes the operation. Decision-making changes it.

Reinforcement learning is a framework for learning such decisions.

Why Decisions Are Different

- Supervised learning learns from a fixed dataset.
- Decision-making selects actions rather than making predictions.
- Those actions influence future states and therefore the data seen later.

Prediction assumes a mostly stationary data source.
Decisions change the state distribution the model will face next.

How Have We Solved Decision Problems?

Rules and expert systems

Encode operational knowledge directly.

Optimization and control

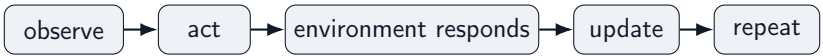
Model the system and compute good actions.

Reinforcement learning

Improve decisions using experience and feedback.

These are not mutually exclusive: practical systems often combine expert knowledge, control, optimization, and learned policies.

The Traditional Reinforcement Learning Paradigm



Classical reinforcement learning:

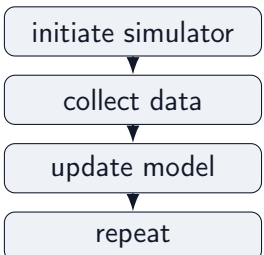
- learns from consequences rather than labeled targets,
- balances exploration and exploitation,
- optimizes long-term reward.

For much of its history, reinforcement learning was framed as learning through continual interaction.

In practice, this usually means learning in a simulator before deployment.

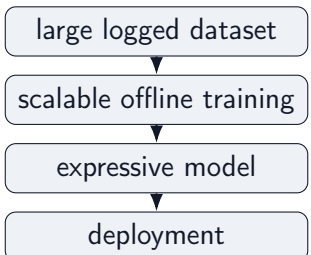
A Data-Centric View of Reinforcement Learning

Interaction-centric RL



Improve through repeated interaction.

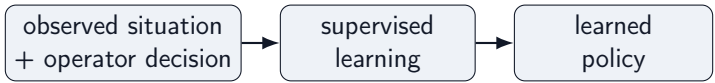
Data-centric ML



Transformers and foundation-model-style scaling benefit from large static datasets.

What can we learn from the interaction data we already have?

Start Simple: Behavior Cloning



- Observe what an operator or existing controller did.
- Train a model to predict the same action in similar situations.
- Deploy the trained model as a learned "policy".

Behavior cloning is fully offline and requires no simulator or reward model.

Can we do more than learn based on operator data? Can we prefer decisions that led to better long-term outcomes?

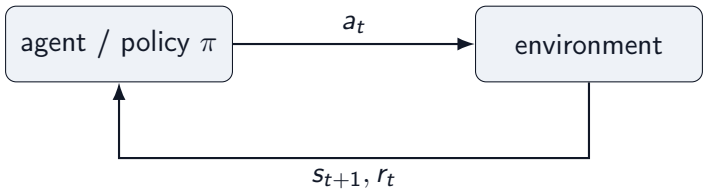
To answer that, we first need a language for decision problems: states, actions, and policies.

States, Actions, and Policies

Object	Grain-terminal example
state s_t	current queue, silo inventory, grain quality, train progress
action a_t	unload truck, choose pit, route conveyor, assign silo, fill car
policy $\pi(a \mid s)$	rule or model that chooses actions from states
next state s_{t+1}	the facility after the action has changed it

A decision problem starts with states, actions, and a policy that maps situations to decisions.

Agent-Environment Loop



- **Agent:** chooses the next action from the current state.
- **Environment:** the facility or simulation that returns the next state and reward.
- **Key contrast:** supervised learning predicts from fixed data; RL actions create the next data.

Imitation Learning: Turn Decisions Into Supervised Learning

First idea: collect observed behavior and train a policy to predict what the operator would do.

Historical demonstrations come from a behavior policy π_b, e.g. human terminal operators:

$$\mathcal{D} = \{(s_t, a_t)\}, \qquad a_t \sim \pi_b(a \mid s_t)$$

Train π_θ as a supervised action predictor:

Continuous actions
steering angle, conveyor speed

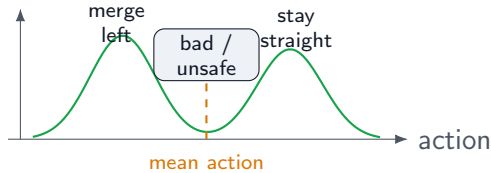
$$\min_\theta \ \mathbb{E}_{\mathcal{D}} \left[\|a - \pi_\theta(s)\|^2 \right]$$

Discrete actions
route choice, silo assignment

$$\min_\theta \ -\mathbb{E}_{\mathcal{D}} \left[\log \pi_\theta(a \mid s) \right]$$

Attractive because it is fully offline, simple to train, and does not require defining a reward.

Pitfall 1: Averaging Demonstrations Can Fail

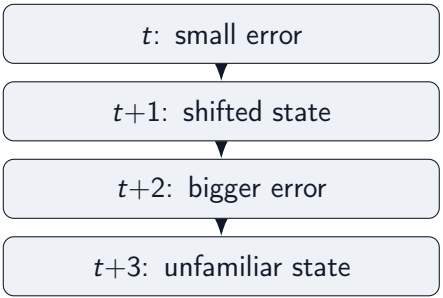


Consider imitation data for self-driving cars:

- in similar observed states, driver 1 stays in lane
- driver 2 changes lanes
- squared-error regression learns the conditional mean action

Problem: the mean action can lie between valid modes: a poor command that no expert demonstrated.

Pitfall 2: Errors Accumulate Over Time



A policy trained only on expert states may not know how to recover from states caused by its own mistakes.

- $\rightarrow$ Distribution shift: deployment samples from the learned policy's state distribution, not the expert's.

Problem: one-step prediction errors can look small while long rollouts become unstable.

Pitfall 3: Imitation Ignores Outcomes

Behavior cloning asks only:

"What action was taken in this situation?"

It does not ask:

"Which action led to the best long-term outcome?"

Every logged decision becomes a training example. The quality of the outcome does not change how strongly it is learned.

Problem: behavior cloning reproduces past decisions, but it cannot prefer decisions with better long-term consequences.

Rewards Measure the Consequences of Decisions

A reward turns delayed product outcomes into feedback for decisions:

$$r_t = r(s_t, a_t, s_{t+1})$$

For grain loading, rewards can combine:

- train loading progress and throughput
- contract quality satisfaction
- penalties for rehandling, congestion, and delay

Rewards tell the learner what outcomes matter, not just what actions were taken.

Long-Term Consequences

A good decision is one that leads to *high future reward*, not just a high immediate reward.

$$\text{Future Reward} = r_t + \gamma r_{t+1} + \gamma^2 r_{t+2} + \cdots$$

where $0 < \gamma < 1$ discounts delayed rewards.
A value function predicts this quantity before making a decision:

$$V(s) \approx \text{future reward, starting from state } s$$

$$Q(s, a) \approx \text{future reward, starting from state } s \text{ and taking action } a$$

Main takeaway

V and Q quantify how good it is to be in a state and to take an action. They are central objects in RL.

Learning the Value Function

Learning the Q-function is an important step in nearly all of RL.

To keep things simple, Q satisfies a recursive relation:
long-term value = expected current reward + discounted future value.

$$Q(s_t, a_t) \approx r_t + \gamma V(s_{t+1})$$

This recursion can be solved with exact or approximate dynamic programming.

We will skip those details and use the equation only as the basic intuition behind value learning.

Three Influential Ideas in Offline Reinforcement Learning

Complementary approaches to improve behavior cloning:

1. **Reweight the logged data**
   Advantage-Weighted Regression (AWR): copy successful decisions more strongly.

2. **Learn conservative value estimates**
   Conservative Q-Learning (CQL): be pessimistic where evidence is weak.

3. **Estimate values only from supported actions**
   Implicit Q-Learning (IQL): avoid maximizing over unsupported actions altogether.

Three Common RL Families

Family	Basic idea
Value-based RL	Learn $Q(s, a)$, act greedily
Policy gradient	Directly improve $\pi(a \mid s)$
Actor-critic	Learn both policy and value

- Q-learning is conceptually simple, but is most natural when actions are discrete.
- Policy gradients apply more generally, but require expensive rollouts.
- Actor-critic methods combine both ideas: they learn a policy and a value estimate, and work with discrete or continuous actions.

Generic Actor-Critic Offline RL

Most modern actor-critic Offline RL methods separate two steps:

$$\begin{array}{c} \text{estimate } Q, V \\ \Updownarrow \\ \text{improve policy} \end{array}$$

The three ideas intervene at different points:

- **AWR** mainly changes the improve-policy step.
- **CQL** and **IQL** focus primarily on value estimation.

Policy updates need values; methods differ in how those values are made reliable.

Problem: Not All Demonstrations Are Equally Useful

Behavior cloning treats every logged decision equally.

But logged data often contains a mixture:

- strong operator decisions,
- reasonable routine decisions,
- choices that led to delay, rehandling, or poor downstream outcomes.

Question

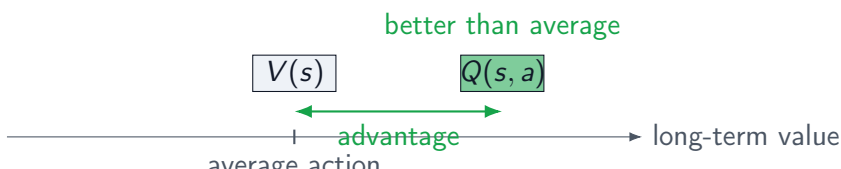
Can we learn more from the better demonstrations?

Core Idea: Advantage

Compare the logged action to what is typical in the same state:

$$A(s, a) = Q(s, a) - V(s).$$

- $Q(s, a)$ measures how good it is to take action a in state s.
- $V(s)$ measures how good the state is on average.
- Their difference says how much better this action is than expected.



This difference is called the **advantage**.

Advantage-Weighted Regression (AWR)

Behavior cloning solves:

$$\min_{\theta} \sum_{i} \ell(\pi_\theta(s_i), a_i)$$

AWR keeps the same supervised-learning loss, but weights each example:

$$\min_{\theta} \sum_{i} \exp(\beta A(s_i, a_i))\, \ell(\pi_\theta(s_i), a_i)\,.$$

Key point

The only change is the weighting. Everything else remains supervised learning.

Next question: Where do reliable estimates of Q and V come from?

Problem: Offline Values Can Be Overconfident

To compute useful weights, we need value estimates.

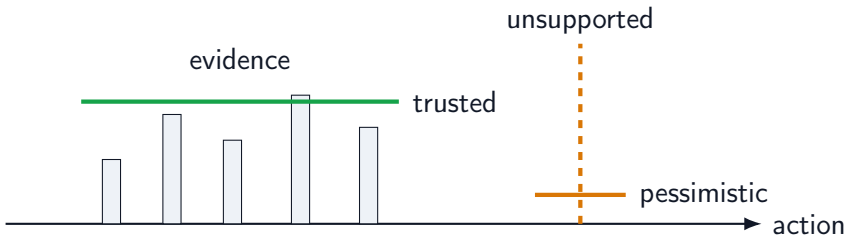
But value estimation is harder offline:

- online learning can test actions and correct bad guesses,
- offline learning cannot collect new feedback,
- unsupported actions may receive overly optimistic predicted values.

Question

How can we learn values without trusting predictions where the data is weak?

Central Idea: Be Pessimistic Where Evidence Is Weak



Conservative value learning lowers confidence away from the logged data.

The policy should not be attracted to actions whose value is high only because the model is guessing.

CQL: Bellman Loss + Conservative Penalty

CQL changes the objective for learning Q:

$$\mathcal{L}_{\mathrm{CQL}} = \underbrace{\mathcal{L}_{\mathrm{Bellman}}}_{\text{fit observed transitions}} + \alpha \left(\underbrace{\mathbb{E}_{a \sim \mu} Q(s,a)}_{\text{push down candidate actions}} - \underbrace{\mathbb{E}_{(s,a) \sim \mathcal{D}} Q(s,a)}_{\text{keep logged actions high}} \right).$$

Intuition

Fit the data, but penalize high values for actions not well supported by the data.

Representative Method: CQL

Conservative Q-Learning changes the value-learning objective.

Its principle is simple:

- fit observed transitions in the dataset;
- penalize high values assigned to actions not well supported by the dataset;
- leave the policy update as a separate step.

What to remember

CQL's contribution is conservative value estimation: do not trust unsupported actions.

Representative algorithm: Conservative Q-Learning (CQL).

Another Design Philosophy: Avoid the Max

Conservative value learning is one way to obtain reliable offline values.

Another family asks whether we can avoid unsupported maximization in the first place.

Many value-based methods rely on searching over actions:

$$\max_a Q(s, a)$$

Offline, this search can expose value errors in weakly supported regions.

Alternative question

Can we estimate values without maximizing over unsupported actions at all?

Problem: Can We Avoid Unsupported Maximization?

Instead of correcting bad value estimates, can we avoid creating them?

Many value-based methods rely on maximizing over actions:

$$\max_{a} Q(s, a)$$

IQL asks a safer question:

What good actions are already supported by the data?

IQL: Replace the Max by a Supported Value

Many value-based methods ask for the best action through:

$$\max_a Q(s, a).$$

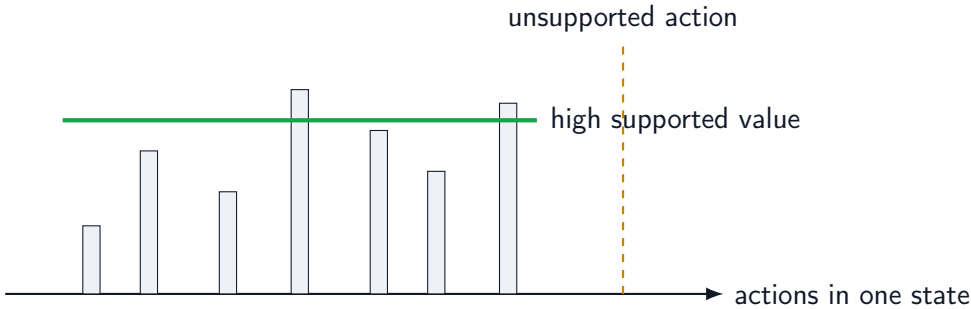
IQL instead estimates:

$$V(s) \approx \text{high expectile of } Q(s, a) \text{ for logged actions } a.$$

Intuition

High but supported, instead of highest possible.

Central Idea: Learn Values Inside the Dataset



IQL estimates high values among logged actions, not over all possible actions.

In practice, this is implemented with expectile regression: a soft way to estimate a high supported value.

Representative Method: IQL

Implicit Q-Learning separates value estimation from action maximization.

- learn values from logged state-action pairs,
- avoid explicit maximization over unsupported actions,
- compute advantages using these in-sample values,
- train the policy with advantage-weighted behavior cloning.

What to remember

IQL = better value estimation + the same policy update as AWR.

IQL Connects Back to AWR

IQL changes how the advantage is computed:

$$A(s, a) = Q(s, a) - V(s).$$

It does not change the policy update:

$$w(s, a) = \exp(\beta A(s, a)), \qquad \min_{\theta} \sum_{i} w(s_i, a_i)\, \ell(\pi_\theta(s_i), a_i).$$

Intuition

IQL gives better weights; the policy update is still weighted behavior cloning.

Remaining Problem: What Should the Policy Look Like?

The first three ideas focus on value estimation.

But policy representation can also matter.

If several distinct actions are good, a simple policy may average them into a poor action.

Optional next question

How should we represent policies when several actions are equally good?

Complementary Issue: Several Good Actions

The first three ideas focused on value estimation.

IDQL addresses a different issue: representing policies when several actions are equally good.

Diffusion Policies Represent Richer Action Distributions

A diffusion policy can sample several plausible actions for the same state.

IDQL combines IQL-style value learning with diffusion-based policy extraction.

Summary: What Changes Across Methods?

Method	Main contribution	Role
AWR	Advantage-weighted policy improvement	Policy update
CQL	Conservative value estimation	Value learning
IQL	In-sample value estimation	Value learning

Main message

Modern Offline RL largely consists of estimating reliable values from logged data, then improving the policy using those values.

Different methods primarily differ in how reliable values are obtained, not in the overall objective.

References

Cheng Chi, Zhenjia Xu, Siyuan Feng, Eric Cousineau, Yilun Du, Benjamin Burchfiel, Russ Tedrake, and Shuran Song.
Diffusion policy: Visuomotor policy learning via action diffusion.
arXiv preprint arXiv:2303.04137, 2023.

Philippe Hansen-Estruch, Ilya Kostrikov, Michael Janner, Jingyun G. Chen, and Sergey Levine.
IDQL: Implicit q-learning as an actor-critic method with diffusion policies.
arXiv preprint arXiv:2304.10573, 2023.

Ilya Kostrikov, Ashvin Nair, and Sergey Levine.
Offline reinforcement learning with implicit q-learning.
arXiv preprint arXiv:2110.06169, 2021.

Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine.
Conservative q-learning for offline reinforcement learning.
In *Advances in Neural Information Processing Systems*, 2020.

Sergey Levine, Aviral Kumar, George Tucker, and Justin Fu.
Offline reinforcement learning: Tutorial, review, and perspectives on open problems.
arXiv preprint arXiv:2005.01643, 2020.

Xue Bin Peng, Aviral Kumar, Grace Zhang, and Sergey Levine.
Advantage-weighted regression: Simple and scalable off-policy reinforcement learning.
arXiv preprint arXiv:1910.00177, 2019.

Richard S. Sutton and Andrew G. Barto.
Reinforcement Learning: An Introduction.
MIT Press, 2 edition, 2018.